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DECISION GOVERNANCE INFRASTRUCTURE

UNITED STATES PATENT AND TRADEMARK OFFICE

PROVISIONAL PATENT APPLICATION

TITLE OF INVENTION:

NON-MARKOVIAN MEMORY KERNEL FOR TIME-SERIES REGIME CLASSIFICATION WITH CONFIDENCE-ADAPTIVE DECISION MAGNITUDE MODULATION

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Applicant / Inventor:	Harold Alberto Nunes Rodelo
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PROVISIONAL APPLICATION FOR PATENT

Application Reference: OMNIX-PAT-2026-002

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Inventor: Harold Alberto Nunes Rodelo

SPECIFICATION

TITLE OF THE INVENTION

NON-MARKOVIAN ADAPTIVE SIGNAL PROCESSING SYSTEM FOR REGIME DETECTION IN AUTOMATED DECISION SYSTEMS WITH OSCILLATORY MEMORY KERNEL AND CONFIDENCE-ADAPTIVE POSITION MODULATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application does not claim priority to any previously filed application. This provisional application is filed pursuant to 35 U.S.C. § 111(b) to establish a priority date. See also related provisional applications OMNIX-PAT-2026-001 (Governance Control Architecture for Automated Decision Systems) and OMNIX-PAT-2026-003 (Ethical and Quantum-Secure Execution Framework), filed concurrently herewith by the same inventor. The Non-Markovian Memory Kernel described herein may operate as an integrated signal source within the governance checkpoint pipeline described in OMNIX-PAT-2026-001.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

Not applicable. This invention was made without Federal sponsorship.

TECHNICAL FIELD

The present invention relates to computer-implemented signal processing methods and systems for time-series data analysis in automated decision environments. More specifically, the invention provides: (1) a non-Markovian memory kernel method that classifies regime states in time-series data by applying an oscillatory exponential decay function to historical data points, weighting recent and periodically reinforced historical events more heavily than intermediate-period events; and (2) a confidence-adaptive position modulation system (CAES) that uses the regime classification and confidence output of the memory kernel to dynamically adjust decision magnitude parameters including position sizing, risk-reward ratios, stop-loss levels, and take-profit targets. Both components are implemented as computational methods integrated within automated decision pipelines.

KEY DEFINITIONS

The following terms are used consistently throughout this specification and the appended claims:

"Non-Markovian": Referring to a computational model in which the current state and future evolution of a system depend on a history of prior states extending beyond the immediately preceding state. A non-Markovian model retains and weights historical information over a configurable memory window, as contrasted with a Markovian model in which the current state depends only on the immediately preceding state.

"Memory Kernel": A weighting function applied to historical time-series data points that assigns a numerical weight to each historical data point based on its temporal distance from the current time point. The memory kernel determines how strongly each historical observation influences the current state computation.

"Oscillatory Decay": A functional form of the memory kernel in which the weight assigned to a historical observation decays exponentially with temporal distance from the current time, but with a periodic oscillatory modulation that reinforces the weights of observations at specific temporal intervals (corresponding to periodically recurring market patterns).

"Regime State": A qualitative classification of the current operating condition of a time-series data source, indicating the type of dynamics currently governing the system's behavior. Regime states in the preferred embodiment include: TRENDING (sustained directional movement), RANGING (oscillation within a bounded price range), VOLATILE (high-amplitude, low-predictability fluctuation), and TRANSITIONAL (intermediate state between two regime types).

"Confidence Score": A numerical value in the range [0.0, 1.0] generated by the memory kernel system indicating the degree of certainty with which the current regime classification can be asserted, based on the consistency and stability of historical observations supporting the classification.

"Aggression Factor": A numerical multiplier computed from the confidence score that modulates the magnitude of an automated decision (e.g., position size, resource allocation) based on the system's confidence in the prevailing regime classification. Higher confidence scores produce higher aggression factors; lower confidence scores produce lower aggression factors.

"Regime Multiplier": A numerical multiplier associated with a specific regime sub-state that adjusts the base decision magnitude to account for the risk-reward characteristics of the identified regime. Different regime sub-states have different expected return and risk profiles, which are reflected in their associated regime multipliers.

"Pipeline Integration": The connection of the memory kernel's output (regime classification and confidence score) to the governance checkpoint pipeline described in OMNIX-PAT-2026-001, wherein the memory kernel's regime signal serves as one of the independent signal sources evaluated by the Decision Contradiction Index (DCI) at CP-6 of said pipeline.

BACKGROUND OF THE INVENTION

1. The Markov Assumption and Its Limitations in Financial Time-Series

Standard computational approaches to time-series regime detection rely on models that incorporate the Markov assumption: the assumption that the current state of a system depends only on the immediately preceding state, and not on any earlier history. Hidden Markov Models (HMMs), first-order autoregressive models (AR(1)), and exponential smoothing methods all embody the Markov assumption to varying degrees.

The Markov assumption is computationally convenient but empirically inadequate for financial time-series data. Financial markets exhibit well-documented non-Markovian characteristics, including:

(a) Long-range dependencies: The behavior of financial markets at a given time is influenced by events that occurred days, weeks, and even months previously. Volatility clustering — the tendency for periods of high volatility to be followed by additional periods of high volatility — is a non-Markovian phenomenon that a first-order Markov model cannot capture.

(b) Periodic memory reinforcement: Financial markets exhibit periodic patterns tied to calendar effects (day-of-week effects, month-end effects, quarterly reporting cycles, annual patterns). These periodic effects cause certain historical time points to be disproportionately informative for predicting current conditions — a phenomenon that a monotonically decaying memory function (such as simple exponential smoothing) cannot represent.

(c) Regime persistence and transition inertia: Financial regime states (trending, ranging, volatile) tend to persist for meaningful durations and to transition between states with characteristic dynamics. Correctly modeling these transitions requires incorporating historical information over a window that extends well beyond the immediately preceding observation.

Existing computational approaches that address non-Markovian behavior in financial time-series include:

- Long Short-Term Memory (LSTM) neural networks, which retain a hidden state vector but do not provide an interpretable memory kernel or a principled confidence score
- Fractional Brownian Motion (fBM) models, which model long-range dependence but do not incorporate oscillatory memory reinforcement or produce regime classifications
- Wavelet decomposition methods, which extract multi-scale features but do not produce a unified regime classification with an associated confidence score

None of the foregoing approaches provides: (1) a closed-form, interpretable, computationally efficient memory kernel that combines exponential decay with periodic oscillatory reinforcement; (2) a regime classification output with a principled confidence score; and (3) direct integration with a confidence-adaptive position modulation system that translates regime uncertainty into calibrated decision magnitude adjustments.

2. The Position Sizing Problem in Automated Decision Systems

A second deficiency of existing automated decision systems is the absence of a principled mechanism for adjusting decision magnitude (position size, resource allocation, commitment level) based on the confidence of the underlying regime classification.

Existing approaches to position sizing in automated financial systems include:

- Fixed fractional sizing (e.g., fixed 1% or 2% of capital per trade), which does not adapt to regime confidence
- Kelly Criterion sizing, which optimizes for expected logarithmic utility but does not incorporate regime state or regime confidence as inputs
- Volatility-scaled sizing (e.g., inverse-volatility weighting), which adjusts for observed volatility but does not incorporate regime classification confidence

None of these approaches provides a mechanism for simultaneously incorporating both (1) the current regime sub-state (e.g., BULL_STRONG vs. BEAR_MODERATE) and (2) the confidence with which that regime sub-state is identified, into a unified position sizing computation. The present invention addresses this deficiency.

SUMMARY OF THE INVENTION

The present invention provides a computer-implemented signal processing system comprising two integrated components:

Component 2A: Non-Markovian Memory Kernel for Regime Detection

The first component is a computational method for classifying regime states in time-series data using a non-Markovian memory kernel with oscillatory decay. The kernel assigns weights to historical data points using a closed-form function that combines exponential decay (capturing the diminishing relevance of older observations) with periodic oscillatory reinforcement (capturing the enhanced relevance of observations at periodically recurring time offsets). The weighted integration of historical data is used to classify the current regime state and generate a confidence score for the classification.

Component 2B: Confidence-Adaptive Entry System (CAES)

The second component is a computational method for modulating automated decision magnitude based on the regime classification and confidence score produced by the Non-Markovian Memory Kernel. The CAES applies a sigmoid-based aggression function to the confidence score to produce an aggression factor, combines this with a regime-specific multiplier corresponding to the identified regime sub-state, and computes a final decision magnitude (position size) as the product of the aggression factor, the regime multiplier, and a base allocation. The CAES additionally adjusts stop-loss levels, take-profit targets, risk-reward ratios, trailing stop parameters, and maximum position percentages as functions of the confidence score and regime sub-state.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a system architecture diagram illustrating the Non-Markovian Adaptive Signal Processing System, showing the relationship between the Time-Series Input Buffer, the Memory Kernel Computation Module, the Regime Classification Engine, the Confidence Score Generator, and the Confidence-Adaptive Entry System (CAES).

FIG. 2 is a graphical representation of the memory kernel weighting function $K(t-s) = \exp(-|t-s| / \tau) \times [1 + \varepsilon \times \cos(\Omega(t-s))]$, illustrating the exponential decay envelope, the oscillatory modulation, and the resulting combined weight as a function of temporal offset (t–s).

FIG. 3 is a process flow diagram illustrating the step-by-step computation from time-series input through regime classification, confidence score generation, and CAES position size computation.

LIST OF FIGURE ELEMENTS

FIG. 1 Elements:

- 210: Time-Series Input Buffer
- 220: Memory Kernel Computation Module
- 221: Decay Parameter τ (tau)
- 222: Oscillation Amplitude ε (epsilon)
- 223: Oscillation Frequency Ω (omega)
- 230: Weighted Integration Engine
- 240: Regime Classification Engine
- 241: Regime State Output (TRENDING / RANGING / VOLATILE / TRANSITIONAL)
- 250: Confidence Score Generator
- 251: Confidence Score Output (0.0 – 1.0)
- 260: On-Chain Data Integration Module (optional)
- 270: Confidence-Adaptive Entry System (CAES)
- 271: Sigmoid Aggression Function
- 272: Regime Sub-State Multiplier Table
- 273: Final Position Size Output
- 274: Composite Parameter Output (stop-loss, take-profit, risk-reward, trailing stop, max position)
- 280: Pipeline Integration Interface (to OMNIX-PAT-2026-001 DCI Module)

FIG. 2 Elements:

- X-axis: Temporal offset (t–s), measured in hours
- Y-axis: Memory kernel weight K(t–s)
- Curve A: Exponential decay envelope $\exp(-|t-s| / \tau)$
- Curve B: Oscillatory modulation $[1 + \varepsilon \times \cos(\Omega(t-s))]$
- Curve C: Combined kernel K(t–s) — product of Curve A and Curve B
- Annotations: τ = 12 hours, ε = 0.35, $\Omega = \pi/6$ rad/hour

FIG. 3 Elements:

- Start node: Raw time-series data input

- Step 310: Load time-series window into Input Buffer (210)
- Step 320: Compute memory kernel weight K(t–s) for each historical point
- Step 330: Apply weighted integration over time-series window
- Step 340: Classify regime state (TRENDING / RANGING / VOLATILE / TRANSITIONAL)
- Step 350: Compute regime sub-state (BULL_STRONG, BEAR_MODERATE, etc.)
- Step 360: Generate confidence score (0.0 – 1.0)
- Step 370: Apply sigmoid aggression function to confidence score
- Step 380: Select regime sub-state multiplier from lookup table
- Step 390: Compute final_position = aggression × multiplier × base_position
- Step 395: Generate composite parameter output
- End node: Decision parameters transmitted to execution pipeline

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

I. SYSTEM OVERVIEW

Referring to FIG. 1, the Non-Markovian Adaptive Signal Processing System comprises a Time-Series Input Buffer (210), a Memory Kernel Computation Module (220), a Weighted Integration Engine (230), a Regime Classification Engine (240), a Confidence Score Generator (250), an optional On-Chain Data Integration Module (260), and a Confidence-Adaptive Entry System (270). The system optionally communicates with the OMNIX Governance Pipeline (OMNIX-PAT-2026-001) through a Pipeline Integration Interface (280).

The system operates as a computational processing pipeline. Raw time-series data (e.g., financial price data, volatility data, on-chain metrics) is received and buffered in the Input Buffer. The Memory Kernel Computation Module applies the non-Markovian kernel weighting function to each historical data point in the buffer. The Weighted Integration Engine aggregates the kernel-weighted data points into a single weighted integral. The Regime Classification Engine maps the weighted integral to a regime state and sub-state. The Confidence Score Generator quantifies the certainty of the regime classification. The CAES translates the regime classification and confidence score into calibrated decision magnitude parameters.

II. MEMORY KERNEL COMPUTATION MODULE (220)

II.A. Kernel Function

The Memory Kernel Computation Module (220) computes a weight value K(t–s) for each historical data point at time s, relative to the current time t, using the following closed-form function:

$$K(t-s) = \exp(-|t-s| / \tau) \times [1 + \varepsilon \times \cos(\Omega(t-s))]$$

where:

- τ **(tau)** is the memory decay constant, controlling the rate at which the influence of historical observations diminishes with temporal distance. A larger τ value causes the kernel to retain historical information over a longer time window; a smaller τ value causes the kernel to decay more rapidly. In one embodiment, τ is calibrated to approximately 12 hours, reflecting empirical calibration against intraday financial market data. In other embodiments, τ may be calibrated to any value appropriate to the deployment domain: for example, τ on the order of hours for intraday financial or energy grid applications, τ on the order of days for supply chain demand forecasting, or τ on the order of minutes for real-time healthcare patient monitoring signals.
- ε **(epsilon)** is the oscillation amplitude parameter, controlling the magnitude of periodic memory reinforcement. A value of $\varepsilon = 0$ reduces the kernel to a simple exponential decay (equivalent to a standard exponential smoothing function). A value of $\varepsilon > 0$ causes the kernel to assign enhanced weight to historical observations at temporal offsets corresponding to multiples of the oscillation period. In one embodiment, ε is calibrated to approximately 0.35, providing

meaningful periodic reinforcement without causing any historical observation to receive a negative weight. In other embodiments, ε may be set to any value in the range (0, 1] appropriate to the amplitude of periodic memory effects present in the target domain.

- **Ω (omega)** is the oscillation frequency parameter, controlling the period of the oscillatory memory reinforcement. The oscillation period is $2\pi/\Omega$. In one embodiment, Ω is calibrated to approximately $\pi/6$ radians per hour, corresponding to an oscillation period of approximately 12 hours, capturing intraday periodicity observed in financial markets. In other embodiments, Ω may be calibrated to capture any domain-relevant periodicity: daily, weekly, or seasonal cycles in supply chain or energy systems; circadian or ultradian rhythms in healthcare applications; or any other periodic structure present in the target time-series domain.

Execution Boundary Integration: The Non-Markovian Memory Kernel and CAES do not themselves make execution decisions. Their output — a regime classification, a confidence score, and a set of calibrated decision magnitude parameters — is submitted to the governance checkpoint pipeline (OMNIX-PAT-2026-001) for evaluation at the execution boundary. A high-confidence regime classification does not independently authorize execution; it constitutes one signal among multiple independent signals that must collectively satisfy admissibility criteria at the execution boundary before any action is permitted to proceed. This design ensures that the regime-adaptive position sizing capabilities of the CAES cannot bypass execution boundary governance.

II.B. Properties of the Kernel Function

The memory kernel $K(t-s)$ has the following mathematical properties that make it suitable for financial regime detection:

- **Non-negativity:** For $\varepsilon \le 1$, $K(t-s) \ge 0$ for all values of $(t-s)$. In the preferred embodiment with $\varepsilon = 0.35$, the minimum value of K is $\exp(-|t-s| / \tau) \times (1 - 0.35) = 0.65 \times \exp(-|t-s| / \tau) > 0$, ensuring that all historical observations receive non-negative weight.
- **Exponential decay:** The factor $\exp(-|t-s| / \tau)$ ensures that the overall magnitude of the kernel weight decreases exponentially with temporal distance, giving priority to recent observations.
- **Periodic reinforcement:** The factor $[1 + \varepsilon \times \cos(\Omega(t-s))]$ causes the kernel weight to be periodically enhanced for historical observations at temporal offsets that are multiples of the oscillation period $2\pi/\Omega$. This models the empirical phenomenon of periodic market memory.
- **Parameter calibratability:** All three parameters $(\tau, \varepsilon, \Omega)$ are configurable, allowing the kernel to be calibrated for different time-series domains (e.g., different financial instruments, different market environments, or non-financial application domains).

Parameter Flexibility: In one embodiment, τ is calibrated to approximately 12 hours, ε is calibrated to approximately 0.35, and Ω is calibrated to approximately $\pi/6$ radians per time unit. These values reflect empirical calibration against financial time-series data. In other embodiments, the parameters may be set to any values appropriate to the deployment domain and data characteristics. For example, in an energy grid demand forecasting application, τ may be calibrated to reflect daily or weekly demand cycles; in a healthcare patient monitoring application, τ may be calibrated to reflect physiological signal periodicities. The functional form of the kernel is preserved across all embodiments regardless of parameter values.

II.C. On-Chain Data Integration

The optional On-Chain Data Integration Module (260) extends the time-series data processed by the memory kernel to include data sourced from distributed blockchain ledger networks, including:

- On-chain trading volume for decentralized exchange pairs
- Net inflow/outflow to centralized exchange wallets
- Active address counts and transaction frequency metrics
- Stablecoin supply growth rates as a proxy for market liquidity

On-chain data is incorporated through a configurable weight parameter that determines the relative contribution of on-chain data versus traditional off-chain market data in the weighted integration computation.

III. REGIME CLASSIFICATION ENGINE (240)

III.A. Primary Regime Classification

The Regime Classification Engine (240) maps the weighted integral produced by the Weighted Integration Engine (230) to one of four primary regime states:

Regime State	Description	Weighted Integral Characteristics
TRENDING	Sustained directional movement in time-series	High weighted integral magnitude with consistent directional sign
RANGING	Oscillation within a bounded range	Low weighted integral magnitude with alternating directional sign
VOLATILE	High-amplitude, low-predictability fluctuation	High weighted integral variance with inconsistent directional pattern
TRANSITIONAL	Intermediate state between regime types	Weighted integral exhibiting declining or increasing trend momentum

III.B. Regime Sub-State Classification

Within each primary regime state, the system further classifies the time-series into one of eleven regime sub-states, each associated with a predetermined regime multiplier used by the CAES:

Regime Sub-State	Regime Multiplier	Description
BULL_STRONG	1.20×	Strong, high-confidence upward trend
BULL_MODERATE	1.10×	Moderate, medium-confidence upward trend
BULL_WEAK	1.00×	Weak, low-confidence upward trend
NEUTRAL_STABLE	0.95×	Stable ranging with low volatility
NEUTRAL_UNCERTAIN	0.85×	Ranging with elevated uncertainty
BEAR_WEAK	0.80×	Weak downward trend
BEAR_MODERATE	0.75×	Moderate downward trend
BEAR_STRONG	0.70×	Strong downward trend — highly defensive posture
VOLATILE_HIGH	0.80×	High-volatility environment — reduced exposure
TRANSITIONAL	0.85×	Regime transition underway — cautious sizing
UNKNOWN	0.70×	Insufficient data for classification — minimum exposure

III.C. Regime Transition Probability

In an alternative embodiment, the Regime Classification Engine additionally computes a regime transition probability, representing the estimated likelihood that the current regime state will change within a defined forward time window. The transition probability is computed from the rate of change of the weighted integral over a sliding temporal window. A high transition probability may be used by downstream systems (including the governance checkpoint pipeline of OMNIX-PAT-2026-001) as an additional signal for evaluating decision admissibility.

IV. CONFIDENCE SCORE GENERATOR (250)

The Confidence Score Generator (250) produces a confidence score in the range [0.0, 1.0] reflecting the degree of certainty with which the current regime classification can be asserted. The confidence score is computed based on:

- **Weighted integral stability:** The consistency of the weighted integral over a trailing stability window. A highly stable weighted integral indicates that the historical data strongly and consistently supports the current regime classification.

- **Kernel weight concentration:** The degree to which the dominant kernel weights are concentrated in recent observations (high concentration → high confidence) versus distributed across a wide historical window (low concentration → lower confidence).
- **Sub-state distinctiveness:** The margin between the score of the identified regime sub-state and the score of the next-closest sub-state. A large margin indicates a clear classification; a small margin indicates ambiguity.

The resulting confidence score is a continuous value that captures the quality of the regime identification, not merely its direction.

V. CONFIDENCE-ADAPTIVE ENTRY SYSTEM (CAES) (270)

Referring to FIG. 3 (Steps 370–395), the Confidence-Adaptive Entry System (CAES) translates the regime classification and confidence score into a complete set of calibrated decision magnitude parameters.

V.A. Sigmoid Aggression Function (271)

The CAES applies a monotone scaling function to the confidence score to produce an aggression factor. In one embodiment, this function is implemented as a sigmoid function. In other embodiments, any strictly monotone increasing function mapping confidence scores in [0, 1] to aggression factors in a bounded positive range may be substituted, including but not limited to: softplus functions, piecewise linear ramp functions, exponential scaling functions, or domain-calibrated lookup tables. The structural requirement — that higher confidence scores produce proportionally larger aggression factors in a smooth and bounded manner — is preserved across all embodiments regardless of the specific functional form chosen.

In the preferred embodiment, the function is implemented as:

aggression = 1 + (2 / (1 + exp(−10 × (confidence − 0.7))))

Properties of this function:

- At confidence = 0.0: aggression ≈ 1.003 (minimum — near-baseline sizing)
- At confidence = 0.7: aggression = 2.0 (threshold — standard sizing)
- At confidence = 1.0: aggression ≈ 2.998 (maximum — fully aggressive sizing)
- The sigmoid shape creates a smooth, differentiable transition between minimum and maximum aggression
- The inflection point at confidence = 0.7 reflects the design choice that meaningfully elevated sizing should occur only when confidence substantially exceeds 50%
- The function is bounded to prevent unbounded position size growth at extreme confidence values

V.B. Final Position Size Computation

The final position size is computed as:

final_position = aggression × regime_multiplier × base_position_size

where:

- **aggression** is the output of the sigmoid aggression function
- **regime_multiplier** is the predetermined multiplier associated with the identified regime sub-state (from the table in Section III.B)
- **base_position_size** is a predetermined baseline allocation, typically expressed as a percentage of total capital under management

The combined effect of the aggression factor (range: approximately 1.0 to 3.0) and the regime multiplier (range: 0.70 to 1.20) produces a final position size range of approximately 0.70 × base (minimum, low confidence + BEAR_STRONG/UNKNOWN regime) to 3.60 × base (maximum, maximum confidence + BULL_STRONG regime). In a preferred embodiment, additional maximum position percentage constraints are applied to prevent the computed position size from exceeding a configurable absolute maximum.

V.C. Composite Parameter Output (274)

In addition to the final position size, the CAES generates a composite parameter output comprising:

Parameter	Computation Basis	Direction
stop_loss_adjustment	Function of regime_multiplier and confidence	Tighter stop at lower multipliers
take_profit_adjustment	Function of regime_multiplier and confidence	Extended target at higher multipliers
risk_reward_ratio	take_profit_adjustment / stop_loss_adjustment	Increases with confidence
trailing_stop_start	Function of confidence	Activates earlier at higher confidence
max_position_percentage	Absolute cap on final_position as % of capital	Hard limit regardless of computed value

VI. PIPELINE INTEGRATION (280)

The memory kernel regime signal (from Regime Classification Engine 240) is transmitted to the Pipeline Integration Interface (280), where it is available as one of the independent signal sources for the Decision Contradiction Index (DCI) module at CP-6 of the governance checkpoint pipeline described in OMNIX-PAT-2026-001. In this integrated configuration, the memory kernel's regime classification is one of at least five independent signals evaluated for pairwise divergence by the DCI. If the memory kernel's regime assessment contradicts the assessments of a sufficient number of other independent signal sources, the DCI score rises toward the CONTRADICTORY threshold, and the proposed action is blocked regardless of the regime multiplier or aggression factor computed by the CAES.

This integration ensures that the adaptive position sizing capabilities of the CAES are never able to override the governance enforcement function of the DCI: a high-confidence BULL_STRONG classification by the memory kernel does not guarantee approval of a proposed action if that classification contradicts the signals of other independent analytical modules.

ALTERNATIVE EMBODIMENTS

Embodiment A: Variable Parameter Calibration by Market Session

In an alternative embodiment, the parameters τ, ε, and Ω are adjusted automatically based on the identified market session (e.g., Asian session, European session, North American session, after-hours). Different market sessions exhibit different memory characteristics, and session-specific kernel calibration may improve regime classification accuracy across session transitions.

Embodiment B: Multi-Asset Kernel Aggregation

In an alternative embodiment, the memory kernel is applied to a plurality of related time-series (e.g., spot price, futures price, implied volatility, and on-chain metrics for a single digital asset) and the resulting weighted integrals are aggregated prior to regime classification. This embodiment captures cross-series dependencies that a single-series kernel would miss.

Embodiment C: Adaptive Oscillation Frequency

In an alternative embodiment, the oscillation frequency parameter Ω is not fixed but is adaptively estimated from the time-series data using spectral analysis (e.g., Fourier transform of the recent time-series). This embodiment allows the kernel to automatically track changes in the dominant periodicity of the market cycle without manual recalibration.

Embodiment D: Bayesian Confidence Updating

In an alternative embodiment, the confidence score is computed using a Bayesian updating framework in which the prior distribution over regime states is updated based on the observed weighted integral. This embodiment provides a probabilistic confidence score with explicit uncertainty bounds.

Embodiment F: Dynamic Kernel Parameter Adjustment

The system may operate with fixed kernel parameters (τ, ε, Ω) calibrated at deployment, with dynamically adjusted parameters updated on a scheduled basis, or with an adaptive structure in which parameters are continuously estimated from the observed time-series data using online learning methods. In all embodiments, the functional form $K(t-s) = \exp(-|t-s|/\tau) \times [1 + \varepsilon \cdot \cos(\Omega(t-s))]$ is preserved, and the regime classification and confidence score outputs maintain the same interface for downstream CAES and pipeline integration regardless of parameter adjustment mechanism.

Embodiment G: Variable Sub-State Count

The system may operate with any number of regime sub-states, from a minimal set of three (bullish, bearish, neutral) to the preferred eleven sub-states or beyond. The CAES regime multiplier table is correspondingly adjusted. The critical structural property — that each sub-state maps to a predetermined multiplier applied to the sigmoid-computed aggression factor — is preserved regardless of the number of sub-states.

Embodiment E: Non-Financial Domain Application

While the preferred embodiment is described in the context of financial time-series, the memory kernel method and CAES are applicable to any time-series domain exhibiting non-Markovian dynamics and periodic memory patterns, including: energy grid demand forecasting, industrial sensor anomaly detection, healthcare patient state monitoring, and supply chain demand signal processing.

EXAMPLES OF USE

Example 1: Cryptocurrency Trading Application

A cryptocurrency trading system uses the Non-Markovian Memory Kernel to process hourly price data for Bitcoin (BTC). With τ = 12 hours, ε = 0.35, and $\Omega = \pi/6$, the kernel assigns enhanced weight to observations from approximately 12 hours ago (corresponding to the previous trading session's close) and from approximately 24 hours ago (corresponding to the same hour of the previous day). The Regime Classification Engine identifies the current regime as BULL_MODERATE with confidence = 0.73. The CAES computes aggression ≈ 2.06, applies the BULL_MODERATE multiplier of 1.10×, and generates a final position size of 2.06 × 1.10 × base = 2.27 × base. This position size is then submitted to the governance checkpoint pipeline, where it is evaluated against risk exposure limits at CP-4 and incorporated into the DCI computation at CP-6.

Example 2: Regime Transition Detection

The Non-Markovian Memory Kernel is monitoring a foreign exchange pair. Over a 6-hour window, the weighted integral transitions from a positive (BULL_MODERATE) value to a near-zero value with high variance. The Regime Classification Engine classifies the current regime as TRANSITIONAL, with confidence = 0.41. The CAES computes aggression ≈ 1.12, applies the TRANSITIONAL multiplier of 0.85×, and generates a final position size of 1.12 × 0.85 × base = 0.95 × base — a near-baseline, defensive sizing appropriate for an uncertain regime transition period.

Example 3: DCI Contradiction Scenario

The memory kernel classifies the current regime as BULL_STRONG with confidence = 0.88, generating aggression ≈ 2.82 and a BULL_STRONG multiplier of 1.20×, for a computed position size of 3.38 × base. However, the Hidden Markov Model simultaneously classifies the regime as VOLATILE_HIGH, and the Kalman filter estimate shows a sharply declining trajectory. These contradictory signals cause the DCI score to rise to 74 (CONTRADICTORY). The governance pipeline at CP-6 issues a BLOCK verdict. The proposed position size of 3.38 × base is never submitted for execution, notwithstanding the memory kernel's high-confidence bullish assessment. This example illustrates the protective function of the DCI integration.

TECHNICAL ADVANTAGES

- **Computational efficiency:** The closed-form kernel function $K(t-s) = \exp(-|t-s| / \tau) \times [1 + \varepsilon \times \cos(\Omega(t-s))]$ is computable in $O(n)$ time for a time-series window of n data points, making it suitable for real-time deployment in high-frequency decision environments.
- **Interpretability:** Unlike LSTM or other neural network approaches to regime detection, the memory kernel has a closed-form mathematical expression with interpretable parameters (τ, ε, Ω), each of which has a clear physical meaning and can be independently calibrated and audited.
- **Principled confidence quantification:** The confidence score generated by the Confidence Score Generator provides a principled, continuous measure of regime classification certainty, enabling the CAES to make smooth, calibrated adjustments to decision magnitude rather than binary switches between fixed position sizes.
- **Governance integration:** The direct integration of the memory kernel's regime signal into the DCI module of the governance checkpoint pipeline (OMNIX-PAT-2026-001) ensures that the regime-adaptive position sizing capabilities of the CAES cannot override the governance enforcement function. High confidence in a bullish regime does not guarantee execution approval; it must be confirmed by the absence of contradiction among other independent signal sources.
- **Domain adaptability:** The memory kernel's three-parameter structure (τ, ε, Ω) allows it to be calibrated for time-series from any domain exhibiting periodic and long-range dependencies, extending its applicability beyond financial markets.

CLAIMS

(Note: Claims are included in this provisional application for reference purposes. Formal claims will be refined in the corresponding nonprovisional application.)

Claim 1 (Broad — Method) — A computer-implemented method for detecting regime states in time-series data, comprising: maintaining a time-series window of data points; computing a memory kernel weight for each historical data point using a function comprising an exponential decay factor and a periodic oscillatory modulation factor; applying said weights to compute a weighted integration of historical data; classifying a current regime state based on said weighted integration; and generating a confidence score for said regime classification.

Claim 2 (Specific — Kernel Formula) — The method of Claim 1, wherein the memory kernel weight for a historical data point at time s relative to current time t is computed according to: $K(t-s) = \exp(-|t-s| / \tau) \times [1 + \varepsilon \times \cos(\Omega(t-s))]$, wherein τ is a memory decay constant, ε is an oscillation amplitude parameter, and Ω is an oscillation frequency parameter.

Claim 3 (Specific — Parameters) — The method of Claim 2, wherein τ is calibrated to approximately 12 hours, ε is calibrated to approximately 0.35, and Ω is calibrated to approximately $\pi/6$ radians per time unit.

Claim 4 (Broad — CAES) — A computer-implemented method for adapting decision magnitude in an automated decision system based on regime classification confidence, comprising: receiving a confidence score from a regime detection system; applying a strictly monotone increasing scaling function to said confidence score to produce an aggression factor, wherein higher confidence scores produce proportionally larger aggression factors within a bounded positive range; applying a regime-specific multiplier corresponding to an identified regime sub-state; and computing a final decision magnitude as a function of said aggression factor, said regime-specific multiplier, and a base allocation value.

Claim 5 (Specific — Sigmoid embodiment) — The method of Claim 4, wherein, in one embodiment, the monotone scaling function is a sigmoid function of the form: $\text{aggression} = 1 + (2 / (1 + \exp(-10 \times (\text{confidence} - 0.7))))$, producing an aggression factor in the range [approximately 1.0, approximately 3.0]. In other embodiments, the monotone scaling function may be any strictly monotone increasing function including softplus, piecewise linear, or exponential functions, provided that higher confidence scores yield larger aggression factors within a bounded range.

Claim 6 (Specific — Sub-State Table) — The method of Claim 4, wherein the regime-specific multiplier is selected from a lookup table comprising at least eleven regime sub-states including BULL_STRONG (1.20×), BULL_MODERATE (1.10×), BULL_WEAK (1.00×), NEUTRAL_STABLE (0.95×), NEUTRAL_UNCERTAIN (0.85×), BEAR_WEAK (0.80×),

BEAR_MODERATE (0.75×), BEAR_STRONG (0.70×), VOLATILE_HIGH (0.80×), TRANSITIONAL (0.85×), and UNKNOWN (0.70×).

Claim 7 (Broad — Integration) — A computer-implemented system wherein a non-Markovian memory kernel regime signal is provided as an independent signal source to a signal contradiction detection module of an automated decision governance pipeline, and wherein said contradiction detection module blocks a proposed action when said memory kernel signal contradicts a threshold number of other independent signal sources.

Claim 8 (Broad — On-Chain) — The method of Claim 1, further comprising integrating on-chain blockchain data from one or more distributed ledger networks into said time-series window with a configurable weight parameter.

Claim 9 (Composite Output) — The method of Claim 4, further comprising generating a composite parameter output comprising a stop-loss adjustment, a take-profit adjustment, a risk-reward ratio, a trailing stop start percentage, and a maximum position percentage, each determined as a function of said confidence score and said regime sub-state.

Claim 10 (Broad System — Maximum Coverage) — A system configured to classify operating conditions in time-series data and to adaptively modulate the magnitude of automated decisions based on said classification, the system comprising one or more processors and a non-transitory computer-readable medium storing instructions that cause the system to: assign weights to historical data points using a function that combines exponential temporal decay with periodic oscillatory modulation; classify a current operating condition based on weighted historical data; generate a confidence score for said classification; and compute a decision magnitude parameter that is a function of both said classification and said confidence score.

Claim 11 (Broad Method — Maximum Coverage) — A computer-implemented method for generating regime-calibrated decision parameters for an automated decision system, comprising: computing a non-Markovian weighted integral of time-series data using a kernel function comprising exponential decay and periodic oscillatory modulation; classifying a regime state from said weighted integral; generating a confidence score for said regime classification; applying a monotonically increasing function of said confidence score to produce a decision magnitude scaling factor; and computing a final decision magnitude as a function of said scaling factor and a regime-specific multiplier.

ABSTRACT

A computer-implemented non-Markovian signal processing system classifies regime states in time-series data by applying a closed-form memory kernel $K(t–s) = \exp(–|t–s| / \tau) \times [1 + \varepsilon \times \cos(\Omega(t–s))]$ to historical data points, combining exponential temporal decay with periodic oscillatory memory reinforcement. The kernel produces a regime classification (TRENDING, RANGING, VOLATILE, TRANSITIONAL) with an associated confidence score. A Confidence-Adaptive Entry System (CAES) translates the regime classification and confidence score into calibrated decision magnitude parameters using a sigmoid aggression function and a regime sub-state multiplier lookup table, producing a final position size and composite parameter output including stop-loss, take-profit, and risk-reward adjustments. The system integrates directly with an automated decision governance pipeline, where the regime signal serves as an independent source for signal contradiction detection. High-confidence regime classifications do not override governance enforcement: contradiction among independent signal sources triggers blocking regardless of regime assessment. The system is applicable to financial trading, digital asset markets, energy, and any time-series domain exhibiting non-Markovian dynamics.

DRAWINGS

FIG. 1 — System Architecture

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■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■        NON-MARKOVIAN ADAPTIVE SIGNAL PROCESSING SYSTEM          ■
■                                                                  ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
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■  ■ TIME-SERIES INPUT    ■    ■ ON-CHAIN DATA INTEGRATION (260)   ■ ■
■  ■ BUFFER (210)         ■    ■ (Optional)                        ■ ■
■  ■                      ■    ■ Blockchain metrics, wallet flows  ■ ■
■  ■ Raw price, volume,   ■    ■ Weight parameter: configurable    ■ ■
■  ■ volatility data      ■    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■                 ■                  ■
■           ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■                  ■
■                             ▼                             ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■  ■       MEMORY KERNEL COMPUTATION MODULE (220)            ■ ■
■  ■                                                          ■ ■
■  ■   K(t−s) = exp(−|t−s| / τ) × [1 + ε × cos(Ω(t−s))]        ■ ■
■  ■                                                          ■ ■
■  ■   τ (221): Memory decay constant ≈ 12 hours              ■ ■
■  ■   ε (222): Oscillation amplitude ≈ 0.35                  ■ ■
■  ■   Ω (223): Oscillation frequency ≈ π/6 rad/hour          ■ ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■ ■
■                         ▼                                 ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■   ■
■  ■ WEIGHTED INTEGRATION ENGINE (230)                      ■   ■
■  ■ Applies K(t−s) weights to time-series data points      ■   ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■   ■
■                      ▼                                    ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■                         ■
■  ■ REGIME CLASSIFICATION ENGINE (240)■                       ■
■  ■                                  ■                       ■
■  ■ Primary: TRENDING / RANGING /    ■                       ■
■  ■         VOLATILE / TRANSITIONAL ■                       ■
■  ■                                  ■                       ■
■  ■ Sub-state: BULL_STRONG /         ■                       ■
■  ■ BULL_MODERATE / NEUTRAL_STABLE / ■                       ■
■  ■ BEAR_STRONG / VOLATILE_HIGH /    ■                       ■
■  ■ TRANSITIONAL / UNKNOWN (11 total)■                       ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■                         ■
■         ■                                                 ■
■         ▼                                                 ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■    ■
■  ■ CONFIDENCE SCORE       ■    ■ CAES (270)                ■    ■
■  ■ GENERATOR (250)        ■■■■■■                            ■    ■
■  ■                        ■    ■ aggression = sigmoid(conf) ■    ■
■  ■ Output: [0.0 – 1.0]    ■    ■ final_pos = agg × mult × base■    ■
■  ■■■■■■■■■■■■■■■■■■■■■■■■■■   ■ + composite param output     ■    ■
■                        ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■    ■
■                                    ■                      ■
■                                    ▼                      ■
■                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■    ■
■                    ■ PIPELINE INTEGRATION (280)        ■    ■
■                    ■ → DCI Module (OMNIX-PAT-2026-001) ■    ■
■                    ■ → Execution Pipeline              ■    ■
■                    ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■    ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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FIG. 2 — Memory Kernel Weight Function

```
Weight
K(t−s)
  ■
1.35■■■■■■■                                            ■■■■■■■
  ■       ■                                          ■
1.00■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  ■           ■               ■■■■■■               ■
0.65■■■■■■■■■■■■■■■■■■■■■■■■■■■   ■■■■■■■■■■■■■■■■
  ■               ■
0.35■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  ■                   ■■■■■■■■■■■■■■■■■■■■■■■■■■■
0.00■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    0    6    12    18    24    30    36    42    48
                  Hours (t−s)

■■■ Curve A: exp(−|t−s|/12) [Decay envelope]
■■■ Curve B: [1 + 0.35·cos(π/6·(t−s))] [Oscillation]
■■■ Curve C: K(t−s) = A × B [Combined kernel]

Oscillation period = 2π/Ω = 2π/(π/6) = 12 hours
Reinforcement peaks at: t−s = 0h, 12h, 24h, 36h...
```

FIG. 3 — CAES Process Flow

```
REGIME CLASSIFICATION + CONFIDENCE SCORE
            ■
            ▼
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Step 370: Apply Sigmoid Function     ■
■                                      ■
■ aggression = 1 + 2/(1+exp(          ■
■             -10×(confidence-0.7)))  ■
■                                      ■
■ conf=0.0 → aggression ≈ 1.00        ■
■ conf=0.7 → aggression = 2.00        ■
■ conf=1.0 → aggression ≈ 3.00        ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                   ■
                   ▼
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Step 380: Select Regime Multiplier   ■
■                                      ■
■ BULL_STRONG     → 1.20×             ■
■ BULL_MODERATE   → 1.10×             ■
■ BULL_WEAK       → 1.00×             ■
■ NEUTRAL_STABLE  → 0.95×             ■
■ NEUTRAL_UNCERT  → 0.85×             ■
■ BEAR_WEAK       → 0.80×             ■
■ BEAR_MODERATE   → 0.75×             ■
■ BEAR_STRONG     → 0.70×             ■
■ VOLATILE_HIGH   → 0.80×             ■
■ TRANSITIONAL    → 0.85×             ■
■ UNKNOWN         → 0.70×             ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                   ■
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```
                   ▼
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Step 390: Compute Final Position     ■
■                                      ■
■ final_position = aggression          ■
■               × regime_multiplier   ■
■               × base_position_size  ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
                   ■
                   ▼
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
■ Step 395: Generate Composite Output  ■
■                                      ■
■ • stop_loss_adjustment               ■
■ • take_profit_adjustment             ■
■ • risk_reward_ratio                  ■
■ • trailing_stop_start                ■
■ • max_position_percentage            ■
■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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End of Specification — Patent Family 2

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Inventor: Harold Alberto Nunes Rodelo

Date: [DATE OF FILING]